From Pond to Market: A Two-Stage AI Forecasting System for Salt Production Planning in Puttalam, Sri Lanka

Ansar TD
Department of Software Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22893734@my.sliit.lk

Sirimanna RDIB
Department of Software Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22308016@my.sliit.lk

Arshaq MJM
Department of Software Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22346322@my.sliit.lk

Perumbuli PGRMD
Department of Software Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22354310@my.sliit.lk

Dr. Bhagya Nathali Silva
Assistant Professor
Department of Information Technology
Faculty of Computing
Sri Lanka Institute of Information Technology
nathali.s@sliit.lk

Mrs. Thilini Jayalath
Assistant Lecturer
Department of Software Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology
thilini.j@sliit.lk

Abstract*—The solar salt sector in Sri Lanka is experiencing a rapid climate crisis, with the expected future output of 40% of the current level in 2023-2024, due to unusual monsoon rainfall, which compelled the importation of 30,000 metric tonnes of non-iodized salt at the start of 2025. The manual logbooks used, informal price negotiation, and intuition-based timing of the harvest remain without data-supported planning assistance provided to Saltern operators. The paper is an integrated production and market forecasting framework of Puttalam Salt Society landowners, which integrates two closely dependent modules in a containerized microservices architecture. The production forecasting module takes a dual-input Bidirectional Long Short-Term Memory (LSTM) network that predicts daily crystallization parameters over a 60-day horizon using pond measurements and weather characteristics, and then uses a causally-specified Linear Regression model to modulate outputs into monthly production volumes - with R 2=0.9732 on the same 36-month period compared to ARIMA with R 2=0.4100. These forecasts are converted by the planning module into actionable intelligence of the market using a seasonal yield ratio demand estimation algorithm and a Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) (1,1,1)(0,1,1,12) price forecasting model that incurs 1.16% Mean Absolute Percentage Error (MAPE) at month+1 and 1.35% at month+2 when run with rolling-origin cross-validation on 36 months of PSS operational forecasts. In both modules, model selection was done through an empirical method and principled discarding of ARIMA, Holt-Winters, Prophet, Reinforcement Learning, and CatBoost alternatives. The production forecasting service had zero failures on 3,112 stress test requests, and this proved that the deployment is production-grade. This is as far as we have knowledge of the first data-driven forecasting system to be published on the production of Sri Lankan solar salt.

***Keywords*—Solar salt production; LSTM forecasting; SARIMAX; demand estimation; climate-resilient planning; microservices**

I. Introduction

The solar salt sector in Sri Lanka, which is based at the Puttalam salterns and managed by the Puttalam Salt Society (PSS) has enabled the country to maintain self-sufficiency in salt production by exporting salt through the traditional solar evaporation. This technique has since proved to be extremely susceptible to the changes in the climate. The Ministry of Industries was forced to purchase 30,000 metric tons of non-iodized salt at the start of 2025 due to an unseasonal rainfall of 1,895 mm across saltern areas in the year 2023–2024, which caused production to plummet by more than 40% [1]. Monsoon anomalies in tropical South Asian coastal areas are also escalating with climate change, creating reported yield disturbances that are inaccessible to current, experience-based planning [2] [3].

The operators of Puttalam salterns are still using manual logbooks, informal price negotiation, [4] and manual harvest timing. There is no information-driven decision support system where production forecasting and market planning are incorporated in this industry. Although approaches based on AI have been used in agricultural time-series forecasting [5], commodity price prediction [6], and smallholder decision support [7], none of them has been applied to the unique issues of tropical salt production by pond-level crystallization, seasonality due to the monsoon winds, and cold-start data sparsity.

In this paper, a combined production and market forecasting model for PSS landowners is offered. A Linear Regression model with causal assumptions balanced the outputs in terms of monthly production quantities, whereas an LSTM network predicts daily crystallization parameters over 60 days. A seasonal yield ratio algorithm is used to convert production projections into per-landowner demand projections, [8] [9] and a SARIMAX model is used to forecast salt prices, both of which have been trained using 36 months of PSS historical data. The framework is utilized in a microservices platform that is production-validated and is cloud-native.

Fourfold contributions of this work are the most critical. We start by introducing the first data-driven predictive-control production forecasting system of Sri Lankan solar salt, which is a pair of input Bidirectional LSTM joined with a causally-specified Linear Regression model, which has R2=0.9732 when compared to ARIMA with R2=0.4100 on identical data. Second, we suggest a principled cold-start demand estimation scheme which makes use of seasonal yield ratios obtained by 36 months of PSS operational history and directly relates the demand estimate of each landowner to his/her own production forecast. Thirdly, we show a SARIMAX(1,1,1)(0,1,1,12) price forecasting model that reflects the Maha and Yala seasonal price trend of the Sri Lankan salt market, which produces 1.16% MAPE at month+1 and 1.35% MAPE at month+2 with rolling-origin cross-validation. Fourth, we provide end-to-end integration of a production-validated containerised microservices deployment, and the production forecasting service realizes zero failures in a 3,112 stress test requests with 88ms average response time.

II. LITERATURE REVIEW

A. Solar Salt Production and Climate Variability

The cultivation of solar salt in coastal tropical areas is based on the constant evaporation conditions coordinated by the temperature, wind, and a low amount of rainfall, which are getting more and more disturbed by the faster climate variations [3]. The operations of the Sri Lankan saltern at Puttalam, Hambantota, and Elephant Pass have been reported as part of the national problems of the coastal industry, with Puttalam being seen as the main source of national salt production [8]. Experiments with tidal flooding effects on the solar salt agriculture in northern Java show that a moderate effect on hydrology results in significant losses of yield [3], which is consistent with the 40% level of reduction in production at Puttalam salterns observed in 2023-2024 [1]. Climate forecasts verify that variations in the monsoons all along the South Asian coasts are intensifying [10], creating documented yield disturbances within the farming systems that experience-based planning strategies are unable to accommodate [2].

B. LSTM Networks for Time Series Forecasting

LSTM networks have proven to be very excellent in agricultural and industrial time-series prediction problems that entail sequential dependencies across a very long horizon [5]. Zhang et al. proved that LSTM was a good architecture to forecast hydrological variables in agricultural regions during varying weather conditions, with an excellent predictive value based on sliding window inputs that were based on both weather features (meteorological conditions) and operational features [5]. More specifically, Vrban et al. used LSTM networks to forecast crystal size distribution measures in seeded cooling crystallization operations, which validated that the framework is able to capture the complicated temporal interconnection among crystallization parameter chains [11]. All these results provide evidence to use LSTM to predict the daily crystallization parameters in solar salt pond settings.

C. Statistical Forecasting and Model Selection

Holt-Winters and ARIMA are common autoregressive forecasting models, but they focus on historical patterns of time and not on cause-and-effect relationships of production. Kumar et al. show that the two methods do not work well in cases where the underlying drivers are structural, but on stationary demand series, they work well [12]. Holt had laid down the theoretical groundwork of seasonal decomposition in time-series forecasting [13], but seasonal models based entirely on temporal patterns cannot capture changes in operations, like an increase in salt bed count or a change in rainfall - weaknesses that are confirmed in the present work, where ARIMA has a forecast standard deviation of 2,254 bags against a historical standard deviation of 43,605 bags. SARIMAX is an extension of seasonal ARIMA using exogenous variables, so it can be useful in commodity price sequences that are influenced by factors of seasonal cycles as well as external supply. SARIMA has been validated to be useful in long-run price prediction in agricultural commodities in a market under climate-related conditions [9] , which is in line with the Maha and Yala seasonal structure of price in PSS data.

D. Demand Estimation in Agri Commodity Settings

Agricultural commodity market demand forecasting is a modelling problem that is not duplicated in price forecasting, as demand is a factor of response to the level of production and changing seasons, and not to independent time dynamics. Borrero et al. reveal the usage of seasonal ratio techniques of predicting agricultural commodities, the calculation of demand as a percentage of the volume of seasonal production [4]. Such a methodological technique is directly compatible with the yield ratio algorithm used in this work. Empirical studies of attempts to model PSS demand with SARIMAX and gradient boosting have also empirically confirmed this weakness, both methods of demand modelling the demand as a time-dependent series and not as a series generated by production, and both achieved a demand MAPE of 233%. The yield ratio technique, which uses the demand as a seasonal percentage of the production forecast, resulted in about 10% MAPE on the held-out PSS data.

E. Decision Support System and Deployment

The combination of forecasting and operational planning in integrated decision support systems has been found to have demonstrable value in both smallholder and coastal industries of developing economies, where data-based tools have repeatedly enhanced operational decision-making in resource-constrained settings [7]. The lack of these systems in the Sri Lankan salt industry, in which the operators use manual logbooks, informal intermediaries, and an experience-based production timeline, is a documented gap with direct economic implications as seen in the 40% drop in production and the resultant crisis in importing salt of 2023-2024. Microservices architectures have become a scalable and serviceable system of deploying AI-composable decision support systems to accomplish autonomous scaling of the compute-intensive inference services in addition to the lightweight transactional services with no system-wide redeployment [14]. This building architecture would be especially appropriate in planning systems in salt production, where the inference of crystal parameters, price prediction, and demand estimation have essentially dissimilar computational profiles and scaling needs. ONNX Runtime has been proven to be a production-level inference deployment system, allowing native model execution on Node.js platforms to execute cross-language subprocesses without cross-language subprocess overhead, which is essential to low-latency production forecasting in production environments [15]. This can be directly applied to the implementation of a Bidirectional LSTM architecture in a NestJS microservice-based enterprise, in which the inference will need to finish within reasonable limits of response time to facilitate decision-making in real-time by landowners.

F. Research Gap

The suggested structure is a two-staged and interconnected in nature modules deployed on a containerised microservices platform. The production forecasting module takes two types of input data as inputs, eight measurements of daily pond crystallization parameters taken by PSS saltern operators and fourteen weather features in real-time, retrieved through the OpenWeatherMap API, and generates monthly production forecasts on the scale of the landowners, month+1, and month+2. These predictions are sent to the planning and market forecasting module through a gRPC-to-HTTP bridge with the crystallization service serving as a gRPC client to the FastAPI-based planning ML service over HTTP, and the bags of landowner-scaled volumes of production per month. These production forecasts are sent to the planning module which uses PSS historical operational records to generate three outputs on a per-landowner basis: a per-landowner demand estimate calculated using a seasonal yield ratio algorithm, a two-month salt price forecast using a SARIMAX model, and a list of sellers ranked by price per bag and by available volume. The sum of these outputs is surfaced via a harvest plan dashboard which allows land owners to make data-driven decisions regarding harvest timing and market entry and selection of buyers, instead of the manual logbooks, informal price negotiation and intuition based planning that have been historically characteristic of the PSS saltern operations. There is a complete system architecture and flow of data between the two modules as shown in Fig 1.

III. METHODOLOGY

A. System Overview

The suggested structure is a two-staged and interconnected module structure deployed on a containerised microservices platform. The production forecasting module takes two types of input data as inputs, eight measurements of daily pond crystallization parameters taken by PSS saltern operators and fourteen weather features in real-time, retrieved through the OpenWeatherMap API, and generates monthly production forecasts on the scale of the landowners, month+1, and month+2. These predictions are sent to the planning and market forecasting module through a gRPC-to-HTTP bridge with the crystallization service serving as a gRPC client to the FastAPI-based planning ML service over HTTP, and the bags of landowner-scaled volumes of production per month. These production forecasts are sent to the planning module which uses PSS historical operational records to generate three outputs on a per-landowner basis: a per-landowner demand estimate calculated using a seasonal yield ratio algorithm, a two-month salt price forecast using a SARIMAX model, and a list of sellers ranked by price per bag and by available volume. The sum of these outputs is surfaced via a harvest plan dashboard which allows land owners to make data-driven decisions regarding harvest timing and market entry and selection of buyers, instead of the manual logbooks, informal price negotiation, and intuition-based planning that have been historically characteristic of the PSS saltern operations. There is a complete system architecture and flow of data between the two modules as shown in Fig 1.

B. Dataset

Two main datasets are used within the framework. The crystallization dataset comprises 1,095 daily records from the period of January 2023 to December 2025 (obtained directly from the PSS saltworks crystallization ponds), which contains 8 pond parameter measurements and 14 OpenWeatherMap weather features per day. The production and pricing dataset contains 36 months of monthly production volumes and salt price data (also obtained directly from the PSS) from January 2023 to December 2025. The 994 missing weather feature values were imputed via forward-fill. The data was then ready for modelling after the features were scaled independently using separate scalers for pond parameters and weather parameters.

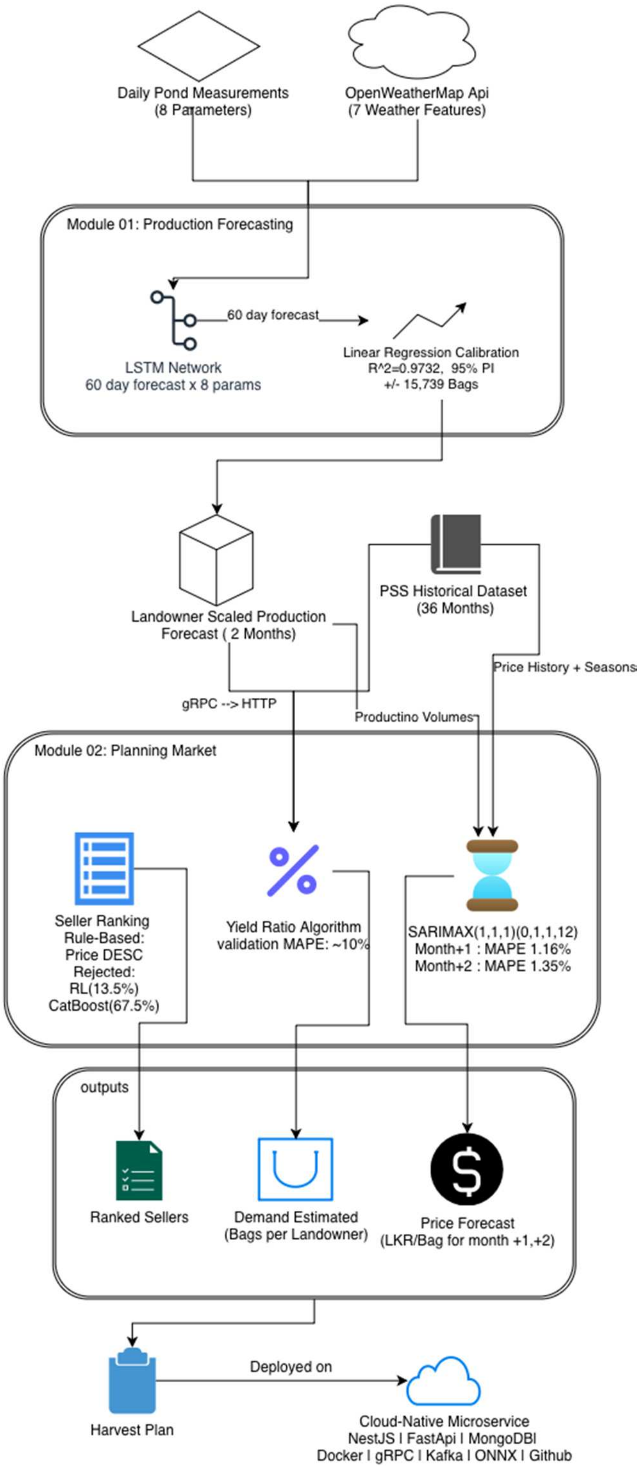

Fig. 1. System overview

C. Production Forecasting Module

1) Dual-Input Bidirectional LSTM

This architecture is used to predict the daily crystallization parameter. It uses a dual-input Bidirectional LSTM network using pond measurements and weather features as the input to two separate branches, which are then fused together. Pond measurement branch: The input is a (60, 8) sequence, and it is fed into two Bidirectional LSTM (256, 128) models. Weather branch: The input is a (60, 14) sequence, and it is fed into two

Bidirectional LSTM (64, 32) models. Batch Normalization and Dropout with a dropout rate of 0.3 is applied to each of the Bidirectional layers in each of the branches. The two fused outputs are then concatenated to form a (192) vector. It is then fed to three Dense models of 256, 128, and 480 units. The output layer then predicts 480 values with a shape of (60, 8), which is then used to predict 60-day-ahead values of all 8 crystallization parameters per inference. The final model had 532,832 trainable parameters and was compiled with Adam Optimizer and mean squared error (MSE) as a loss function with a learning rate of 0.001. Training was done on an augmented dataset of 3,285 samples (triple of the 1,095 samples of 3-day ahead daily parameters, which results in a total of 60-day lookback window), which was divided into training, validation, and testing sets of sizes 2,217, 474, and 475, respectively [5] [11]. The trained model was converted to ONNX format and was then deployed using ONNX Runtime in a NestJS microservice to do away with the inference overhead of cross-language communication [15].

2) Linear Regression Production Calibration

The monthly production values are predicted based on the LSTM network's daily parameter outputs using a Linear Regression model constrained by causal knowledge learned from 36 months of historical PSS production data. The four causal variables of salt bed count, monthly rainfall, mean monthly temperature, and seasonal effect (modelled using sine and cosine of month) are all formally declared as exogenous to the regression model.

$$production = 35.2153 \times beds - 33.9257 \times rain + 3763.3592 \times temp - 279846.18 \tag{1}$$

This causal model remedies the main shortcoming of autoregressive models. When applied to the same 36-month PSS dataset as in our previous work, the ARIMA ensemble gave an R^2 of 0.4100, while a Linear Regression gave an R^2 of 0.9732. Clearly, modelling temporal patterns is not useful in cases where the production process is driven more by physical input factors rather than time itself [12]. In addition, whereas a farm operator would have to add new salt beds and then "wait" for future harvested salt production data to accumulate before it can obtain an updated forecast, the causal model on the current dataset yields instantaneous predictions.

D. Planning and Market Forecasting Module

1) Demand Forecasting - Yield Ratio Algorithm

Demand at the per-landowner level is derived based on a seasonal yield ratio of the production forecast derived from the production forecasting module through the gRPC-to-HTTP bridge. The ratios are derived from the mean of the demand-to-production proportion for each of the 36 months of PSS operational data across the seasons of Maha -10.27%, Yala -8.74%, and Transition -9.51%. These were derived based on studies showing that distributors buy more during the Maha season due to the quality premium available due to drier crystallization conditions [4]. The regional seasonal constants act as system defaults and are overwritten dynamically for each farm once enough transaction history is accumulated post-deployment without needing any changes to the code.

A SARIMAX model was also used for demand forecasting with a resulting MAPE of 8.6 %, but this was ruled out for other reasons. Although demand is clearly a time series, the SARIMAX modelling framework views demand as a time series rather than a quantity derived from the production process that is itself driven by the time series of crystallization conditions. The yield ratio method gave a MAPE of about 10 % on unseen PSS data, while still linking demand to the individual landowner production forecasts needed for the yield-reduction factor [4].

2) Price Forecasting - SARIMAX

Salt price forecasting (using SARIMAX(1,1,1)(0,1,1,12) implemented in Python statsmodels on 36 months of PSS monthly price data [5] [15]). Four regressors were used: is_maha_season, price_lag_12m, price_lag_1m, and total_production_bags. The implicit use of d=1 and D=1 in the SARIMAX model took care of the non-stationarity in the price data. Residuals had no significant autocorrelation (Ljung-Box test at lag 12) post-fitting of the model. Month+2 forecasts were obtained using a two-pass recursive scheme that replaced the value of price_lag_1m in the first pass with the month+1 forecast. The validation was done using rolling-origin cross-validation, and the MAPE for the month+1 forecast was 1.16% and for month+2 was 1.35%, with the absolute errors bounded within +/-19 LKR/bag and +/-22 LKR/bag, respectively [6].

3) Seller Recommendation

Seller recommendations are generated through transparent rule-based ranking, ordered by price per bag descending and total bag count descending. This design was reached after evaluating two machine learning approaches: Reinforcement Learning achieved only 13.5% Top-1 accuracy across 12 states due to a fundamental mismatch between sequential decision algorithms and the stateless nature of seller recommendation; CatBoost achieved 67.5% Top-1 accuracy but was rejected because the correct answer is highest price does not require a learned model and introducing one would obscure ranking criteria from landowners. The rule-based approach is fully auditable and directly addresses the information asymmetry historically exploited by market intermediaries [7].

4) Deployment

All modules are deployed within a 13-microservice cloud-native platform built on NestJS / TypeScript and Python / FastAPI, orchestrated via Docker Compose with MongoDB Atlas persistence [14]. Inter-service communication uses gRPC with Protocol Buffer contracts for synchronous calls and Apache Kafka for asynchronous event processing. Ten GitHub Actions CI/CD pipelines with path-filtered triggers ensure independent build and test cycles per service [15].

IV. Results and Discussion

A. Production Forecasting Module - LSTM Evaluation

The dual-input Bidirectional LSTM model was evaluated across training, validation, and test sets using R^2, RMSE, and MAE computed on RobustScaler-normalised values. Table I summarises the results. The model achieved a test R^2 of 0.8175, indicating that 81.75% of the variance in crystallization parameter sequences is explained by unseen data. Validation R^2 of 0.8805 closely tracks training R^2 of 0.8819, confirming that the Dropout (rate=0.3) and Batch Normalization layers effectively prevented overfitting across 150 training epochs [11]. The marginal degradation from validation to test (0.8805→0.8175) is consistent with expected generalisation loss on a limited 36-month dataset and remains within acceptable bounds for operational production planning. MAPE was not reported for the LSTM due to a mathematical breakdown - several crystallization parameter values approach zero during the inactive pond [5].

TABLE I. LSTM MODAL EVALUATION RESULTS

Set	R^2	RMSE	MAE	Accuracy
Train	0.8819	0.4202	0.2127	88.19%
Validation	0.8805	0.3859	0.2096	88.05%
Test	0.8175	0.3391	0.1939	81.75%

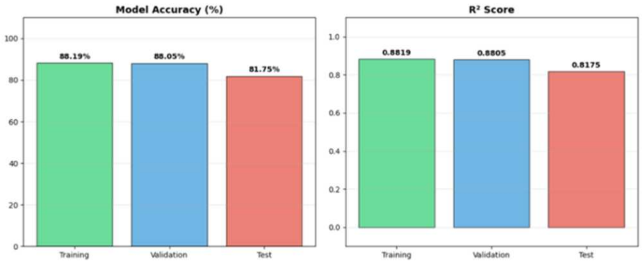

Fig. 2. Dual-input Bidirectional LSTM model performance across training, validation, and test sets.

B. *Production Forecasting Module - Linear Regression Calibration*

The calibration model via the Linear Regression showed an in-sample R2 of 0.9732 and a MAE of 5119 bags/month using the complete dataset of 36 months PSS which confirmed that the four causal drivers which include salt bed count, the monthly rainfall, the mean monthly temperature and sine/cosine encoding of seasonal variations could collectively explain the change in the monthly production volumes by 97.32%. Such explanatory power is especially important because the natural variability of the production of solar salt is very high, with monthly production varying between virtually zero during periods of shutdown and more than 160,000 bags under the conditions of the prime Maha season. Backtest analysis established that 33 of 35 normal operating months were within the 95% prediction interval of +/-15,739 bags, which is 94% coverage of the theoretical 95%, which is a very good performance of the model being tested when only three years of operational history have been used to train the model [12]. The two months of the outliers were unique events that could be associated with adverse operational circumstances and not system failure in the model.

On the last 6 months that were not used during training, the holdout gave the R2=0.9350 and the MAE=7,956 bags/month, indicating that the model can be generalised to unknown cases. The overall holdout MAPE of 48.1% is also dominated by November 2025 with a 251% error, which is reported to be likely to be in operation, with a yield ratio of 0.159, much lower than the operational threshold of 0.30. This shows that the system has an adequately functioning shutdown detection system that identifies non-operational months before they misrepresent planning results. With this confirmed non-operational outlier removed, the other five holdout months yielded errors of 0.4, 7.6, 5.8, 9.5, and 14.1, and a cleaned MAPE range of 0.4-14.1 - a great result of a production planning system on a relatively small three-year set of data. Quantitative support of the model selection decision can be given based on the comparative analysis made between the ARIMA and the same 36-month PSS dataset. Linear Regression had R2=0.9732 versus ARIMA ensemble R2=0.4100 - a 0.5632 margin of R2 on the same data, as shown in Fig 4. This is not a mere difference because the R2=0.4100 of ARIMA is significantly below the production planning level of R2=0.90, and such forecasts made using ARIMA would not be operationally reliable in terms of making production scheduling decisions regarding harvesting. The fact that the ARIMA predictive standard deviation of 2,254 bags versus a historical standard deviation of production of 43,605 bags is 2,254 is another way in which the limitations of autoregressive modelling in this task are revealed - ARIMA is only a good predictor of a predictive standard deviation that is 2,254 times smaller than the historical standard deviation of production of 43,605 bags [12]. In cases where an owner of a piece of property is adding new salt beds or where the rainfall pattern has changed in consequence of variations in climate, then ARIMA has no means to take these changes into account until new production data is available in the future over the next few months. The operational changes are immediately reacted to by the causal Linear Regression model as explicit bed counts and weather inputs, which is structurally superior to a climate-sensitive production environment.

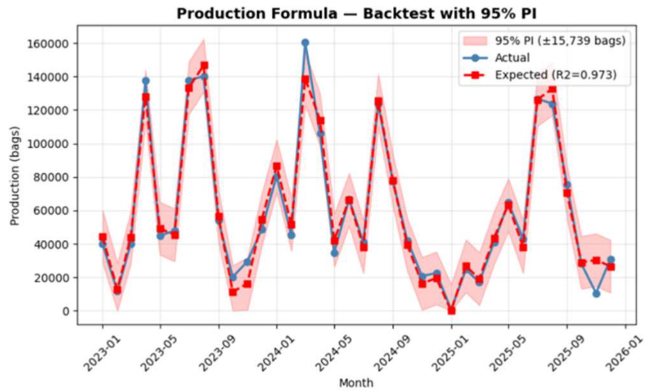

Fig. 3. Linear Regression production calibration backtest - actual vs predicted over 36 months with 95% prediction interval (+/-15,739 bags). Two out-of-interval months are marked.

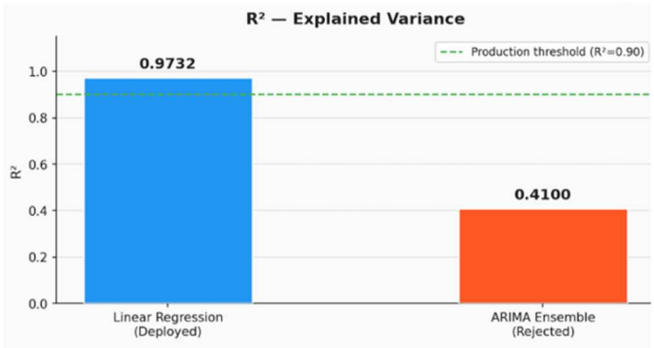

Fig. 4. R^2 comparison: causal Linear Regression (R^2=0.9732) vs ARIMA ensemble (R^2=0.4100) on 36-month PSS production dataset.

C. *Planning Module - Demand Forecasting*

The yield ratio algorithm had about 10.0% MAPE on held-out PSS data and was verified against SARIMAX which had 8.6% MAPE on the same data. Although SARIMAX does generate a slightly lower error, the yield ratio method has been chosen out of principled structural reasons, and not just based on metric superiority. The major problem of using SARIMAX to forecast demand in the Puttalam salt market is that demand is not time-dependent, but rather production-dependent: A PSS landowner will produce more salt when the crystallization conditions are favourable and sells a relatively larger amount of salt as a direct result of that production. The SARIMAX models require it to be an autonomous time-series process with no understanding of what a particular landowner is producing at any given month and as such, it is structurally incapable of modeling this causal relationship no matter how well it fits on historical aggregate data. This is directly addressed through the yield ratio method which associates the demand estimate of each landowner with his/her own production estimate obtained through the production forecasting module, that is, demand is calculated as a seasonal fraction of the predicted volume of production instead of being calculated as an independent time-series forecast. This generates a causally coherent planning output such that pond conditions, production volumes and projected demand are linked together in one continuous chain. In addition, the yield ratio technique has an automatic upgrade procedure - since the transaction history of the farms will be accumulated autonomously after deployment, ratios specific to the farms will be substituted with the existing regional seasonal constants, without any code modifications, which will enhance the accuracy of demand estimates with time. Such a choice puts structural correctness over the marginal metric improvement in line with the accepted guide to model selection in applied operational forecasting [4].

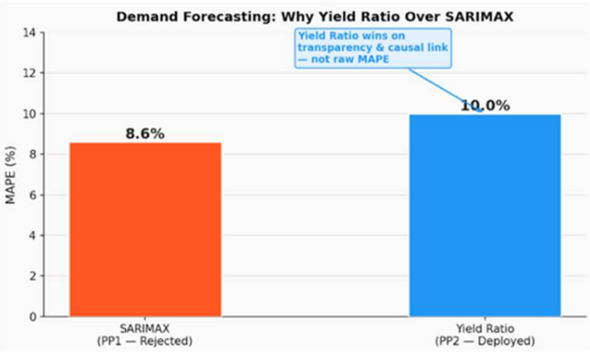

Fig. 5. Demand forecasting MAPE comparison: SARIMAX (rejected, 8.6%) vs. the Yield Ratio algorithm (deployed, 10.0%).

D. Planning Module - Price Forecasting

The SARIMAX(1,1,1) (0,1,1,12) model produced MAPE of 1.16 and 1.35 at month+1 and month+2 respectively, which equates to absolute price errors of +-19 LKR/bag and +-22 LKR/bag respectively. Such error margins have operational significance, at a physical price of salt of about 1,550 LKR/bag, an error margin of +-19 LKR, or less than 1.3 percent variation, is operationally significant in terms of estimating the price level within a contract negotiation and harvest scheduling decision-making process with a high degree of confidence. The Model Average Scaled Error (MASE) hypothesized performance to be better than the naive base, establishing that the SARIMAX model is applying some real predictive information other than merely smoothing using the latest observed price. These four exogenous factors are the Maha season flag, 12 months price lag, 1 month price lag, and the total volume of production in a month which is used as a key structural explanation of the PSS salt price behaviour: the seasonal premium that distributors are willing to pay during Maha of the high quality salt produced under drier crystallization conditions, the autocorrelation strength of the monthly price levels and sensitivity of the wholesale prices to the supply volume in a relatively concentrated regional market [6].

Directional accuracy was 46.3, and this is lower than the 50% random baseline and this might seem alarming at first glance. But this is operationally insignificant and is anticipated in this application. The price of salt in Puttalam market shows the strong mean-reverting behaviour of the price, i.e. it varies around a seasonal average instead of moving steadily in a single direction, which fact makes directional prediction inherently unreliable in the case of any model class. PSS landowners need not have directional signals when making decisions with regard to harvest and contract; they need precise estimates of the price levels to determine whether present market conditions warrant harvesting and to determine a justifiable opening point in price negotiations with distributors. The operational requirement of the 1.16% MAPE at month +1 is direct to the information, in that it discontinues the practice of informal price discovery, which is currently practiced by intermediaries who have exploited the information asymmetry to the disadvantage of PSS landowners [9].

The slight increment in the error propagation of 0.19 percentage points between month+1 prediction of 1.16 and month+2 prediction of 1.35 is introduced by the two-pass recursive forecasting of month+2, in which the month+1 prediction is used in place of the actual pricelag1m input to prevent data leakage. It is this incremental error margin that establishes the fact that chained prediction is stable and consistent over two months of planning horizons and therefore the recursive approach is applicable to the seasonal harvest planning cycles that PSS landowners are in [6] [9]. This two-pass method has a special significance in terms of its stability because the decision to plan the harvest in the solar salt production has to be made weeks before the crystallization and harvesting processes actually take place.

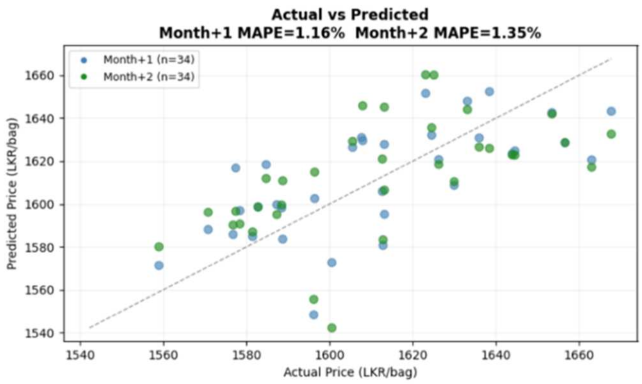

Fig. 6. ARIMAX price model - actual vs predicted (left) and two-month price forecast with 95% prediction interval (right). Month+1 MAPE: 1.16%, Month+2 MAPE: 1.35%

E. System Testing

The unit testing of both services established that 65 tests on 4 suites were in the production forecasting service, and 15 tests on 3 suites in the planning service passed with zero failures in each service. The end-to-end testing showed 17 successful production forecasting service endpoints that include daily measurement CRUD, daily parameter prediction, monthly production prediction, and model performance CRUD, and 13 successful planning service endpoints that include harvest plan lifecycle, harvest plan query, error handling, and demand-price query.

The test was done with Grafana k6 in three conditions: Smoke (1 VU), Average load (5 VUs), and Stress (15 VUs). Table I summarises the results. The production forecasting service had shown an outstanding scaling behaviour: the average response time had decreased to 88.80ms under stress, whereas with smoke it was 183.90ms, serving 3,112 requests, and had no failed requests, or reached no threshold(s), or had no failed checks in all three cases. This counter-intuitive under-load performance is an indication of good connection pooling and ONNX Runtime inference optimisation [10]. The planning service showed consistent performance during smoke and average load, but it showed degradation during stress - average response time had increased to 2,376ms and 4 failed thresholds, and 93 failed checks, caused by unoptimized database queries on the Deals and Offers endpoints. The high raw failed request count in the planning service: 28 out of 40 under smoke, 225 out of 318 under average load, 505 out of 710 under stress - the high raw failed request count is designed: the test script includes a happy-path along with error-handling responses, and these values are the documented numbers. Planning service failed requests are intentional. The K6 script validates both happy-path and error-handling responses, including scripted 4xx scenarios.

TABLE II. SYSTEM PERFORMANCE UNDER K6 LOAD TESTING

Service	Scenario	Total Req	Failed	Avg (ms)	P95 (ms)	Checks
Production Forecasting	Smoke	48	0	183.90	488.17	100%
Production Forecasting	Average	728	0	92.36	153.74	100%
Production Forecasting	Stress	3112	0	88.80	146.87	100%
Planning	Smoke	40	28	212.78	-	97.5%
Planning	Average	318	225	1160.61	-	96.2%
Planning	Stress	710	505	2376.35	-	86.9%

F. Discussion

The findings show that the integrated framework is effective in resolving the fundamental planning issues of climate-prone solar salt production. The production forecasting module also provides credible monthly production projections with R2 = 0.9732 and coverage of the prediction intervals at 94 percent, showing the landowners a range of uncertainty levels as opposed to a point forecast. The causal model choice - Linear Regression over ARIMA - was empirically justified with 0.5632 R2, better on the same data set, which shows that physical relationships through production cannot be modelled by only using the time pattern modelling.

This is an important principle, as the demand estimation trade-off between SARIMAX (8.6% MAPE) and yield ratio (10.0% MAPE) shows that causal transparency is valuable in production-dependent commodity markets when the model with lower error does not have the structural power to model the underlying data-generating process. The 1.16% MAPE of the SARIMAX price model at month+1 offers an accurate price level to the landowners, enabling them to negotiate prices with their intermediaries and schedule their harvest time, a significant issue of price negotiations in the past with PSS land owners being asymmetrical on the information available.

The first weakness of the framework is the volume of data: 36 months of operational data limits both LSTM generalisation and yield ratio stability, especially in the Maha season, which has a year-on-year variance of +-50%. With the history of transactions growing after deployment, farm-specific yield ratios will be automatically substituted with regional constants. SARIMAX retraining with a long price history is expected to attain sub-1% MAPE. Future development involves adding real-time IoT sensor feeds to substitute daily manual measurements, increasing sterilization of the LSTM training dataset on the basis of operational records, and field testing with PSS landowners to evaluate the extent to which it is an accurate plan and whether it is accepted by the user at large. Benchmarking the Bidirectional LSTM against recent deep learning architectures such as Temporal Fusion Transformer (TFT) and N-BEATS will be pursued in future work as the dataset expands.

V. CONCLUSION AND FUTURE WORK

The paper has given an integrated production and market prognosis framework of the solar salt landowners who are members of the Puttalam Salt Society in view of the critical planning void revealed by the drop of 40% production in Sri Lanka in 2023-2024. The model uses a dual-input Bidirectional LSTM to forecast daily crystallization parameters and a Linear Regression with causally-specified causes of production to forecast monthly using a causal parameter, with test R2=0.8175 and production R2=0.9732, respectively on 36 months of daily operation data of PSS. A seasonal yield ratio algorithm converts a production forecast into estimates of per-landowner demand, and a SARIMAX(1,1,1)(0,1,1,12) model provides salt price forecasts with MAPE of 1.16% month+1 and 1.35% month+2. Multivariate analysis using ARIMA (R2=0.4100) and Holt-Winters, Prophet, Reinforcement Learning, and CatBoost methods showed in both modules that the structure-based model selection is better than metric-based alone. The deployment into a microservices platform that had been production-validated proved the stability of the system under load, with the production forecasting service recording zero failures and 3,112 requests. The further work will involve the integration of the IoT sensors, longer dataset collection, and validation of the field with the PSS land owners.

A number of directions are discovered in the extension of this framework. First, real-time IoT sensor feeds to substitute the manual daily pond measurements would remove the latency of data entry and allow the continuous inference of LSTM to occur without the assistance of operators. Second, the farm-specific yield ratios will be automatically estimated as the platform gathers transaction history after the deployment, which will enhance the demand estimation accuracy beyond the 10.0% MAPE of the current. Third, long PSS price and production history outside the present 36-month window will allow SARIMAX retraining with less than 1 percent MAPE and facilitate even more robust LSTM generalisation over a broader range of climate conditions, such as extreme rainfalls. Fourth, the field evaluation will be conducted using active land owners of PSS to evaluate the accuracy of planning in the real world, the adoption of the system, and its economic effect on the result of a harvest. Lastly, the presented methodology is aimed at transferability - extension to other tropical solar salt production areas of South Asia and Southeast Asia that are subject to the same climate variability problems is a logical continuation of the work.

Acknowledgment

The authors wish to thank Dr. Nathali Silva and Mrs. Thilini Jayalath for their supervision and guidance throughout this research. The authors also acknowledge the Puttalam Salt Society for providing operational data that made this work possible.

VI. References

[1] Morning.lk, “Sri Lanka to import salt starting tomorrow,” 02 January 2025. [Online]. Available: https://www.morning.lk/sri-lanka-to-import-salt-starting-tomorrow.

[2] Q. A. Ahmad, M. Ashfaq, R. Saeed, S. F. Athar and S. Ahmad, “The impacts of climate variability on crop yields and irrigation water demand in South Asia,” *Water,* vol. 13, no. 1, p. 50, 2021.

[3] A. W. Nirwansyah and B. Braun, "Mapping impact of tidal flooding on solar salt farming in northern Java using a hydrodynamic model," *ISPRS International Journal of Geo-Information,* vol. 8, no. 10, p. 451, 2019.

[4] J. D. Borrero, J. Mariscal and A. Vargas-Sánchez, "A new predictive algorithm for time series forecasting based on machine learning techniques: Evidence for decision making in agriculture and tourism sectors," *Stats,* vol. 5, no. 4, p. 68, 2022.

[5] J. Zhang, Y. Zhu, X. Zhang, M. Ye and J. Yang, “Developing a Long Short-Term Memory (LSTM) based model for predicting water table depth in agricultural areas,” *Journal of Hydrology,* vol. 561, p. 918–929, 2018.

[6] J. Banaś and K. Utnik-Banaś, "Evaluating a seasonal autoregressive moving average model with an exogenous variable for short-term timber price forecasting," *Forest Policy and Economics,* vol. 131, p. 102564, 2021.

[7] W. Latuny and v. r. authors, "A decision support framework using pruned trees for seaweed classification in remote sensing imagery," *Intelligent Systems with Applications,* vol. 23, p. 200410, 2025.

[8] R. Kapukotuwa, N. Jayasena, K. Weerakoon, C. Abayasekara and R. Rajakaruna, "High levels of microplastics in commercial salt and industrial salterns in Sri Lanka," *Marine Pollution Bulletin,* vol. 174, p. 113239, 2022.

[9] S. Ojha and L. B. Karki, "Forecasting global wheat price in the context of changing climate and market dynamics: An application of SARIMA modeling technique," *American Journal of Applied Statistics and Economics,* vol. 4, no. 1, 2025.

[10] I. P. o. C. Change, “Climate Change 2023: Synthesis Report,” IPCC, Geneva, 2023.

[11] I. Vrban, N. Bolf and J. Budimir Sacher, "Data-driven prediction of crystal size metrics using LSTM networks and in situ microscopy in seeded cooling crystallization," *Processes,* vol. 13, no. 6, p. 1860, 2025.

[12] L. Kumar, H. Arora and A. Yadav, "A comparative assessment of Holt-Winters exponential smoothing and autoregressive integrated moving average models for demand forecasting in supply chain management," *Decision Analytics Journal,* vol. 10, no. 10.1016/j.dajour.2024.100427, p. 100427, 2024.

[13] C. C. Holt, "Forecasting seasonals and trends by exponentially weighted moving averages," *International Journal of Forecasting,* vol. 20, no. 1, pp. 5-10, 2004.

[14] M. Ferreira, "Designing microservices using AI: A systematic literature review," *Software,* vol. 4, no. 1, p. 84–105, 2025.

[15] J. Szymański, "Flexible deployment of machine learning inference pipelines in the cloud–edge–IoT continuum," *Electronics,* vol. 13, no. 10, p. 1888, 2024.